

DATA ENGINEERING

Distributed Systems, Scalable Processing, and Modern Architectures

FIRST EDITION · 2026

PART I The Data Engineering Landscape

PART II Distributed File Systems

PART III Batch Processing at Scale

PART IV In-Memory Processing

PART V Data Pipelines & Integration

PART VI Insight Delivery

PART VII The Modern Data Stack

Hadoop

HDFS

MapReduce

Apache Spark

Kafka

Delta Lake

HiveQL

MLOps

Data Mesh

Apache Beam

Lakehouse

Diff. Privacy

Rokan Uddin Faruqui

Professor, Department of Computer Science & Engineering
University of Chittagong